

MATHEMATICAL STATISTICS 202
ASSIGNMENT – TERM 4

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Question 1

(a) Capture the data into R

```
degree <- c("BA", "BSc", "BCom", "BCom", "BA", "BA", "BA",  
            "BCom", "BA", "BCom", "BCom", "BA", "BA", "BSc")  
  
save(degree, file = "23G4435_question1.Rdata")
```

(b) Numerical and graphical summaries

```
degree  
  BA BCom  BSc  
   7   5   2  
> prop.table(table(degree))  
degree  
      BA      BCom      BSc  
0.5000000 0.3571429 0.1428571  
> |
```

Degree of Students Using the Lib

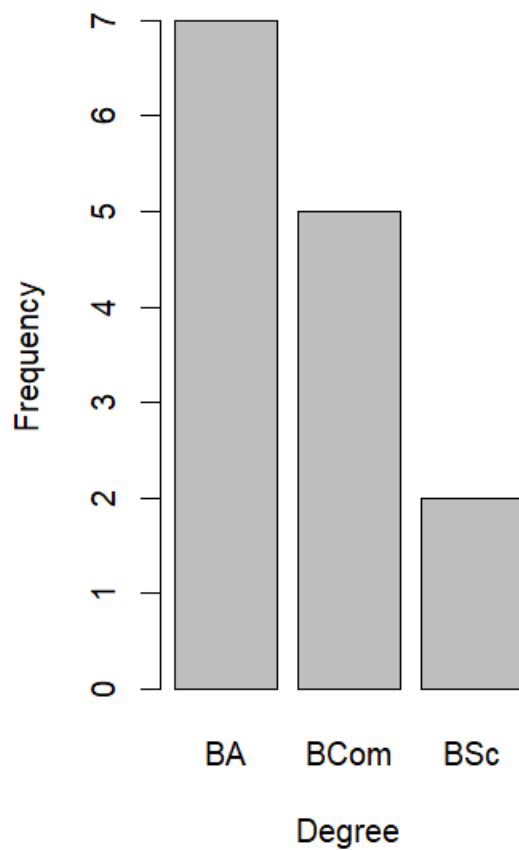


Figure 1. Bar chart of degree frequencies.

Question 2

a)

```
# QUESTION 2
q2 <- read.csv("23G4435/23G4435_question2.csv")
q2$Location <- factor(q2$Location)
q2$Response <- factor(q2$Response,
                      levels = c("Strongly Disagree",
                                "Disagree",
                                "Uncertain",
                                "Agree",
                                "Strongly Agree"))
```

(b) Graphical and numerical summaries

```
> # Numerical summaries
> table(q2$Location)
```

```
non-Rhodes    Rhodes
      543      613
```

```
> table(q2$Response)
```

```
Strongly Disagree    Disagree    Uncertain    Agree
      218      194      331      244
Strongly Agree
      169
```

```
> q2_table <- table(q2$Location, q2$Response)
```

```
> q2_table
```

	Strongly Disagree	Disagree	Uncertain	Agree	Strongly Agree
non-Rhodes	101	55	218	123	46
Rhodes	117	139	113	121	123

Graphical summary

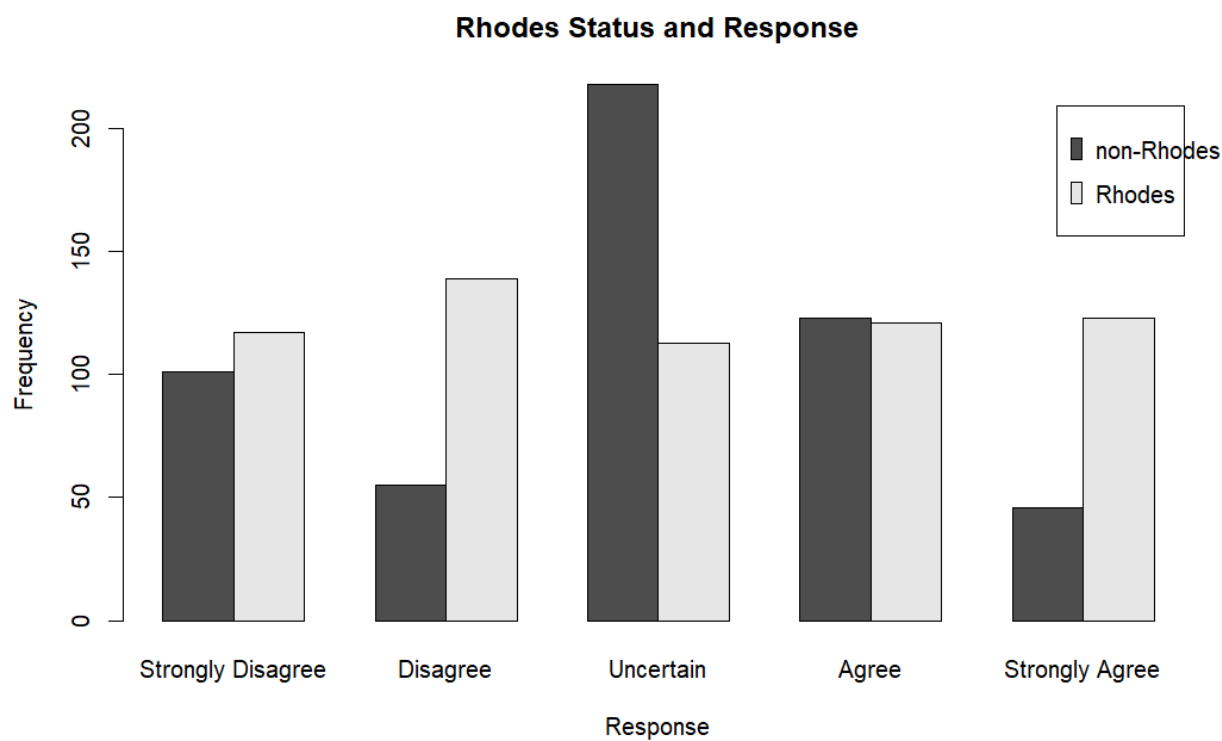


Figure 2. Responses by Rhodes/non-Rhodes status.

(c) Decision and Conclusion.

```
> q2_chisq <- chisq.test(q2_table)
> q2_chisq
```

Pearson's Chi-squared test

```
data: q2_table
X-squared = 102.09, df = 4, p-value < 2.2e-16
```

```
> q2_chisq$expected
```

	Strongly Disagree	Disagree	Uncertain	Agree
non-Rhodes	102.3997	91.1263	155.4784	114.6125
Rhodes	115.6003	102.8737	175.5216	129.3875

	Strongly Agree
non-Rhodes	79.38322
Rhodes	89.61678

A chi-square test of independence is appropriate because both variables are categorical. The expected cell counts are all comfortably above 5, so the chi-square approximation is appropriate.

Using the Pearson chi square test, Decision at $\alpha = 0.05$: reject H_0 of independence.

Conclusion: The data provides sufficient evidence of a relationship between respondents' Rhodes/non-Rhodes status and their response to the statement about Rhodes University as a brand.

Question 3

```
# QUESTION 3
q3 <- read.csv("23G4435/23G4435_question3.csv")
a) q3
```

(b) Graphical and numerical summaries

```
# Numerical summaries
by(q3$Fecundity, q3$Zone, summary)
aggregate(Fecundity ~ Zone, data = q3,
  FUN = function(x) c(n = length(x),
    mean = mean(x),
    sd = sd(x),
    median = median(x)))
```

	Zone	Fecundity.n	Fecundity.mean	Fecundity.sd
1	HigherLittorinid	18.00000	84.66667	12.57448
2	Mussel	10.00000	53.20000	10.05319

	Fecundity.median
1	83.00000
2	54.00000

```
# Graphical summary
boxplot(Fecundity ~ Zone, data = q3,
        main = "Fecundity by Tidal Zone",
        xlab = "Tidal zone",
        ylab = "Egg production")
```

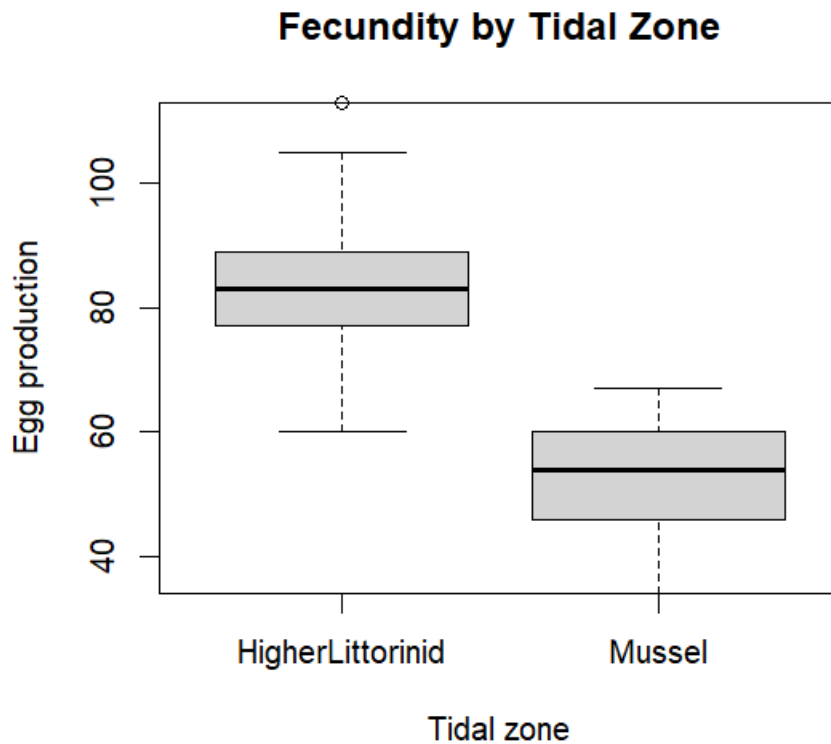


Figure 3. Fecundity by tidal zone.

(c) Decision and Conclusion

```
# Two-sample t-test with equal variances
q3_test <- t.test(Fecundity ~ Zone, data = q3,
                  var.equal = TRUE)
q3_test
```

Results:

Two Sample t-test

```
data: Fecundity by Zone
t = 6.7825, df = 26, p-value = 3.376e-07
alternative hypothesis: true difference in means between group HigherLittorinid and group Mussel is not equal to 0
95 percent confidence interval:
 21.93023 41.00311
sample estimates:
mean in group HigherLittorinid
      84.66667
mean in group Mussel
      53.20000
```

Decision at $\alpha = 0.05$ we reject H_0 .

Conclusion: The data provide sufficient evidence that average egg production differs between the two tidal zones. The observed mean fecundity is higher in the Higher Littorinid zone (84.67) than in the Mussel zone (53.20).

Question 4

```
# QUESTION 4
q4 <- read.csv("23G4435/23G4435_question4.csv")
a) q4$Sex <- factor(q4$Sex)
```

(b) Graphical and numerical summaries

```
# Numerical summaries
by(q4$MetabolicRate, q4$Sex, summary)
aggregate(MetabolicRate ~ Sex, data = q4,
          FUN = function(x) c(n = length(x),
                              mean = mean(x),
                              sd = sd(x),
                              median = median(x)))
```

	Sex	MetabolicRate.n	MetabolicRate.mean	MetabolicRate.sd
1	Female	9.00000	1417.11111	29.30633
2	Male	15.00000	1115.73333	11.53545
	MetabolicRate.median			
1	1417.00000			
2	1118.00000			

```
# Graphical summary
boxplot(MetabolicRate ~ Sex, data = q4,
        main = "Metabolic Rate by Sex",
        xlab = "Sex",
        ylab = "Metabolic rate (kJ)")
```

Results

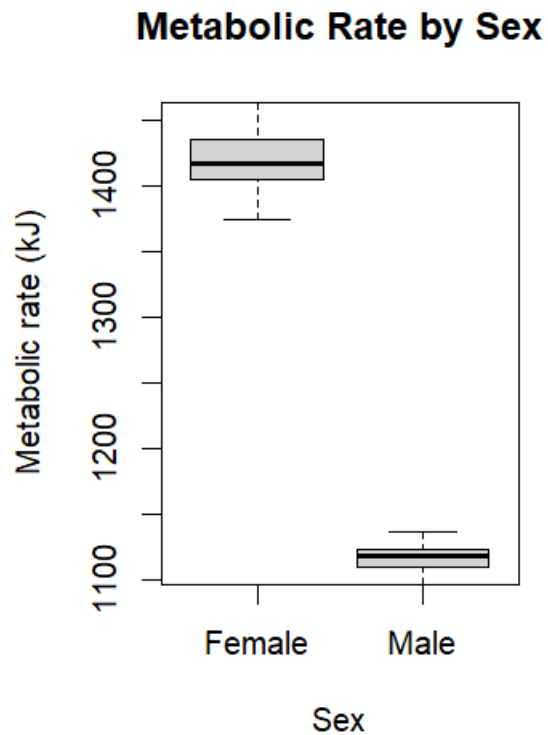


Figure 4. Metabolic rate by sex.

(c) Decision and Conclusion.

```
# Check equality of variances
fligner.test(MetabolicRate ~ Sex, data = q4)
# Welch two-sample t-test
q4_test <- t.test(MetabolicRate ~ Sex, data = q4,
                  var.equal = FALSE)
q4_test
```

Results:

Welch Two Sample t-test

```
data: MetabolicRate by Sex
t = 29.51, df = 9.5095, p-value = 1.13e-10
alternative hypothesis: true difference in means between group Female and
group Male is not equal to 0
95 percent confidence interval:
 278.4623 324.2933
sample estimates:
mean in group Female    mean in group Male
      1417.111           1115.733
```

Decision at $\alpha = 0.05$ we reject H_0 .

Conclusion: The data provide sufficient evidence that average metabolic rates differ between male and female breeding northern fulmars. The observed mean metabolic rate is higher for females (1417.11 kJ) than males (1115.73 kJ).

Question 5

(a) Graphical and numerical summaries

```
# Numerical summaries
summary(q5$P24Level, q5$Dose, summary)
aggregate(P24Level ~ Dose, data = q5,
          FUN = function(x) c(n = length(x),
                              mean = mean(x),
                              sd = sd(x),
                              median = median(x)))
```

Results:

```
q5$Dose: 300
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 10.00  17.75   23.00   24.46  28.00   54.00
```

```
-----
q5$Dose: 600
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 19.00  41.50   67.00   64.62  86.50  102.00
```

	Dose	P24Level.n	P24Level.mean	P24Level.sd	P24Level.median
1	300	24.000000	24.458333	9.748932	23.000000
2	600	8.000000	64.625000	30.170646	67.000000

```
# Graphical summary
boxplot(P24Level ~ Dose, data = q5,
        main = "p24 Level by AZT Dose",
        xlab = "Dose (mg)",
        ylab = "p24 level")
```

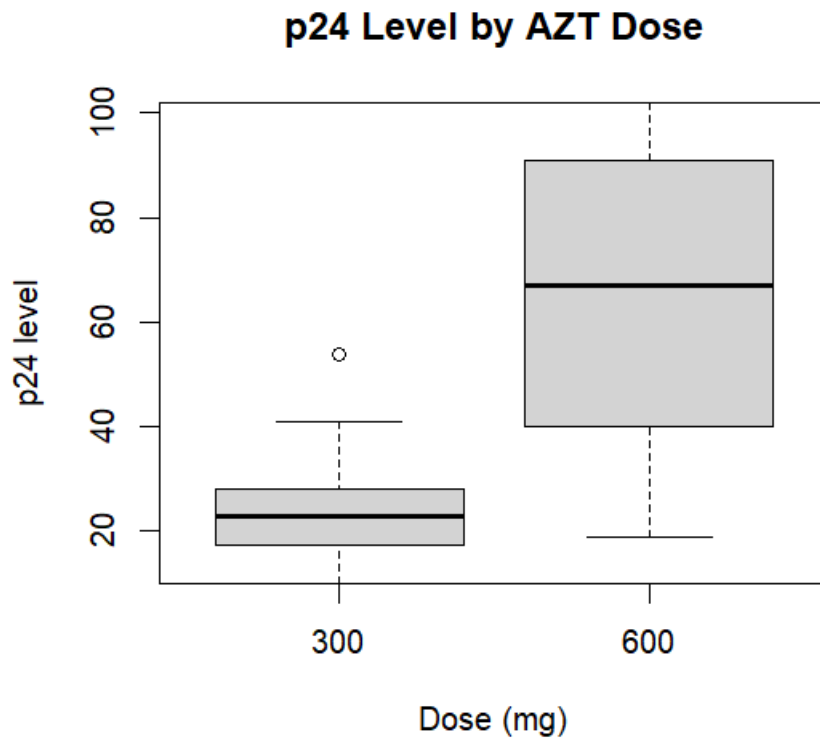


Figure 5. p24 levels by AZT dosage.

(b) Decision and conclusion.

```
# Check equality of variances
fligner.test(P24Level ~ Dose, data = q5)

# Welch two-sample t-test
j5_test <- t.test(P24Level ~ Dose, data = q5,
                  var.equal = FALSE)
j5_test
```

Results:

```
welch Two Sample t-test

data: P24Level by Dose
t = -3.7017, df = 7.493, p-value = 0.006767
alternative hypothesis: true difference in means between group 300 and group 600 is not equal to 0
95 percent confidence interval:
 -65.48686 -14.84647
sample estimates:
mean in group 300 mean in group 600
    24.45833      64.62500
```

Decision at $\alpha = 0.05$: reject H_0 .

Conclusion: The data provide sufficient evidence that average p24 levels differ between the 300 mg and 600 mg dosage groups. The observed mean p24 level is higher for the 600 mg group (64.63) than for the 300 mg group (24.46).

Question 6

```
q6 <- read.csv("23G4435/23G4435_question6.csv")
q6
a) q6_complete <- q6[complete.cases(q6$Lab1, q6$Lab2), ]
```

(b) Graphical and numerical summaries

```
# Numerical summaries
summary(q6_complete$Lab1)
summary(q6_complete$Lab2)
mean(q6_complete$Lab1)
sd(q6_complete$Lab1)
mean(q6_complete$Lab2)
sd(q6_complete$Lab2)
```

Results:

```
> # Numerical summaries
> summary(q6_complete$Lab1)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 389.9  414.4   417.4   417.4  423.3   429.1
> summary(q6_complete$Lab2)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 459.9  473.1   480.9   482.2  491.4   505.6
> mean(q6_complete$Lab1)
[1] 417.4328
> sd(q6_complete$Lab1)
[1] 9.198313
> mean(q6_complete$Lab2)
[1] 482.2239
> sd(q6_complete$Lab2)
[1] 12.2857

# Paired graphical summary
plot(q6_complete$Lab1, q6_complete$Lab2,
     xlab = "Lab 1",
     ylab = "Lab 2",
     main = "Lab 1 vs Lab 2")
abline(0, 1)
```

Results:

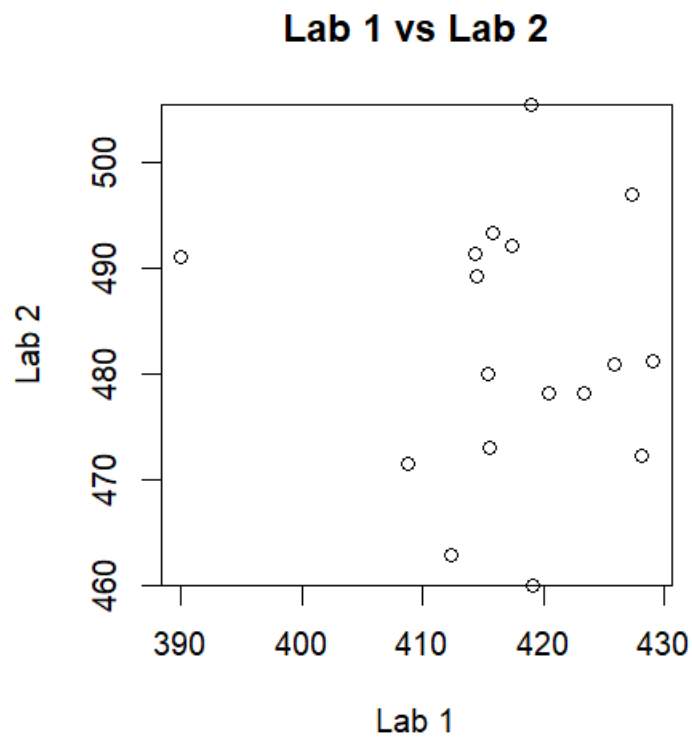


Figure 6. Paired laboratory measurements for the 17 complete samples.

(c) Decision and Conclusion

```
# Differences
difference <- q6_complete$Lab1 - q6_complete$Lab2
summary(difference)
mean(difference)
sd(difference)

# Paired t-test
q6_test <- t.test(q6_complete$Lab1,
                  q6_complete$Lab2,
                  paired = TRUE)

q6_test
```

Results:

Paired t-test

```
data: q6_complete$Lab1 and q6_complete$Lab2
t = -16.88, df = 16, p-value = 1.285e-11
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -72.92805 -56.65417
sample estimates:
mean difference
 -64.79111
```

Decision at $\alpha = 0.05$: reject H_0 .

Conclusion: The data provide sufficient evidence of a significant difference between the average biomass-to-chlorophyll ratios measured by the two laboratories. The observed Lab 2 mean (482.40) is higher than the Lab 1 mean (417.43).

Question 7

QUESTION 7

```
q7 <- read.csv("23G4435/23G4435_question7.csv")
q7$Treatment <- factor(q7$Treatment,
                      levels = c("Control",
                                "Treatment 1",
                                "Treatment 2"))
```

a)

(b) Graphical and numerical summaries

```
# Numerical summaries
by(q7$Yield, q7$Treatment, summary)
aggregate(Yield ~ Treatment, data = q7,
          FUN = function(x) c(n = length(x),
                              mean = mean(x),
                              sd = sd(x),
                              median = median(x)))
```

Results:

```

> # Numerical summaries
> by(q7$Yield, q7$Treatment, summary)
q7$Treatment: Control
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 383.0  394.0   404.7   402.4  409.5   425.7
-----
q7$Treatment: Treatment 1
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 314.1  319.3   321.7   324.4  329.7   340.8
-----
q7$Treatment: Treatment 2
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 369.2  384.5   394.1   393.2  400.6   413.8
> aggregate(Yield ~ Treatment, data = q7,
+           FUN = function(x) c(n = length(x),
+                               mean = mean(x),
+                               sd = sd(x),
+                               median = median(x)))
  Treatment Yield.n Yield.mean Yield.sd Yield.median
1   Control    12.000000 402.387500 12.315106  404.735000
2 Treatment 1    10.000000 324.350000  8.360339  321.725000
3 Treatment 2    24.000000 393.202083 12.350474  394.110000
.

# Graphical summary
boxplot(Yield ~ Treatment, data = q7,
        main = "Fruit Yield by Treatment",
        xlab = "Treatment",
        ylab = "Yield (kg)")

```

Results:

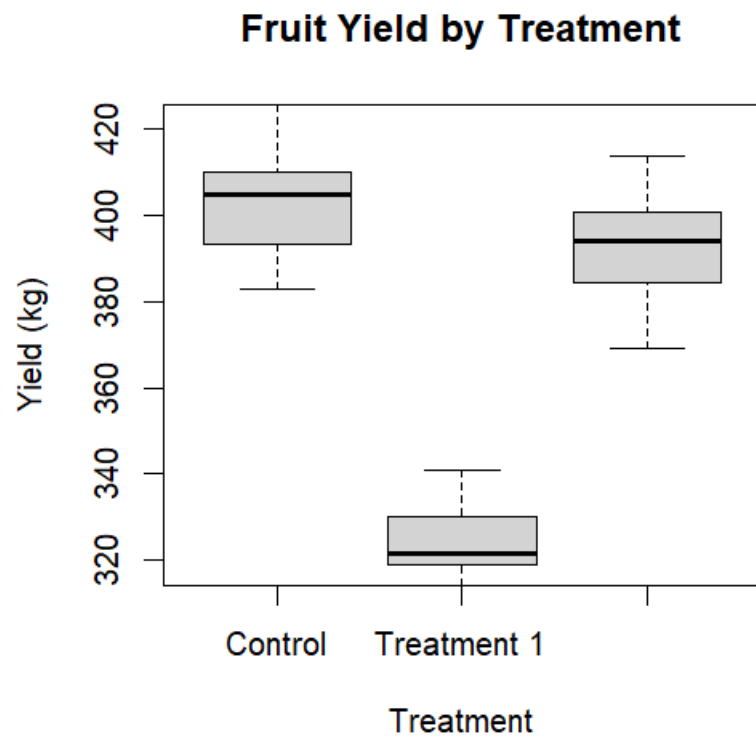


Figure 7. Fruit yield by treatment.

(c) Decision and Conclusion

A one-way ANOVA is appropriate because the response variable (yield) is quantitative and there are three independent treatment groups. The ANOVA assumptions were also checked in R using a variance check and a Shapiro-Wilk test on the residuals; the residual normality check did not indicate a departure from normality.

```
> q7_model <- aov(Yield ~ Treatment, data = q7)
> summary(q7_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Treatment	2	41148	20574	152.4	<2e-16 ***
Residuals	43	5806	135		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Decision at $\alpha = 0.05$ we reject H_0 that all three population means are equal.

Conclusion: The data provide sufficient evidence that average fruit yield differs among the three conditions.